PH.45130 **FALL 2026**

Quantum Gravitation

Problem Set 3

Yuran Seo Due 2026-09-20 23:59

Topic

Canonical Quantization

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Canonical Quantization. State all assumptions and conventions.

$$[q_{ks}, p_{k's'}] = i\hbar\delta_{kk'}\delta_{ss'}$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$H = \sum_{k,s} \hbar\omega_{ks}(a^{\dagger}_{ks}a_{ks} + 1/2)$$

3. Verify the third relation in two limiting regimes and explain the physical meaning of the result.

$$D_{ij,kl}(x) = \langle 0|T h_{ij}(x) h_{kl}(0)|0\rangle$$

Show enough intermediate work that another student can reproduce the calculation.